Circles and first previews

Lesson questions

20 daily questions include C1 and C2. Use separate paper when helpful.

P1. A circle has diameter 8 cm. Find its circumference in terms of π and using 3.14.

P2. A circle has radius 3 cm. Find its diameter and circumference in terms of π.

P3. Find the area of a circle with radius 4 cm. Give an exact answer and an approximation using 3.14.

P4. A circle has diameter 6 cm. Find its area in terms of π and using 3.14.

P5. Circle A has diameter 4 cm and circumference 4π cm. Circle B has diameter 10 cm and circumference 10π cm. Compare C ÷ d.

P6. A collection grows from 80 cards to 100. Find the percent increase. Then find the percent decrease from 100 back to 80.

P7. Start with 80 tokens. Increase by 25%, then decrease the result by 25%. Find the final amount. Explain why it is not 80.

P8. Use steps on a number line for −2 + 5, −3 + (−4), and 2 − 6.

F. A circle has diameter 16 cm. Complete: radius = __; area = π × __² = __π square cm.

D. Maya finds area from a diameter of 14 cm by writing π × 14² = 196π. Explain and repair it.

X1. A circle has diameter 8 cm. Find circumference in terms of π and using 3.14.

X2. A circle has radius 3 cm. Find area in terms of π and using 3.14.

X3. A circle has diameter 8 cm. Find radius and exact area.

X4. A count rises from 80 to 100. Find the increase and percent increase.

X5. Start with 120 points. Increase by 10%, then decrease the result by 10%. Find the final amount.

X6. On a number line, start at −4, move right 7, then left 5. Find the final number and total distance.

S. For circle area, I first identify __. A percent change compares the change with __.

E. A circle has diameter 12 cm. Find its area in terms of π and using 3.14. Show the radius you used.

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Final mixed review - assessment alternative

20 daily questions include C1 and C2. Use separate paper when helpful.

Use this in place of ordinary practice, or in a separate assessment sitting. It is not added to the 20 daily questions.

1. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

2. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

3. A machine makes 24 labels in 6 minutes. Find labels per minute and minutes per label.

4. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

5. Find $9 \div 40$.

6. Expand $2(x + 7)$.

7. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

8. Find the opposite of -6.

9. Find Q1, Q3, and IQR for 0, 2, 3, 5, 7, 9, 10, 12. Exclude the median if the count is odd.

10. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

Worked example - compare with the slide model

Find area when a circle’s diameter is 10 m.

The radius is half the diameter: 5 m.

Square the radius: $5^2 = 25$.

Area = 25π square m, or about 78.5 square m using 3.14.

Result: 25π square m; about 78.5 square m

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Challenge and reasoning

20 daily questions include C1 and C2. Use separate paper when helpful.

C1. A circular pool has diameter 10 m. A square deck is 14 m on each side, with the pool cut out of its center. Find the deck’s area in terms of π and using 3.14. Then find the pool’s boundary length. Explain why these answers have different units.

C2. Wheel A has radius 20 cm. Wheel B has twice that radius. Find each circumference and area in terms of π. If both roll the same distance without slipping, how many turns does B make while A makes 12? Explain.